Real-Time Video Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

Shunyang Li | Public technical brief | October 2026



Agenda

01 Toward Real-Time H3 Generation
02 vLLM-Omni and ComfyUI Applications

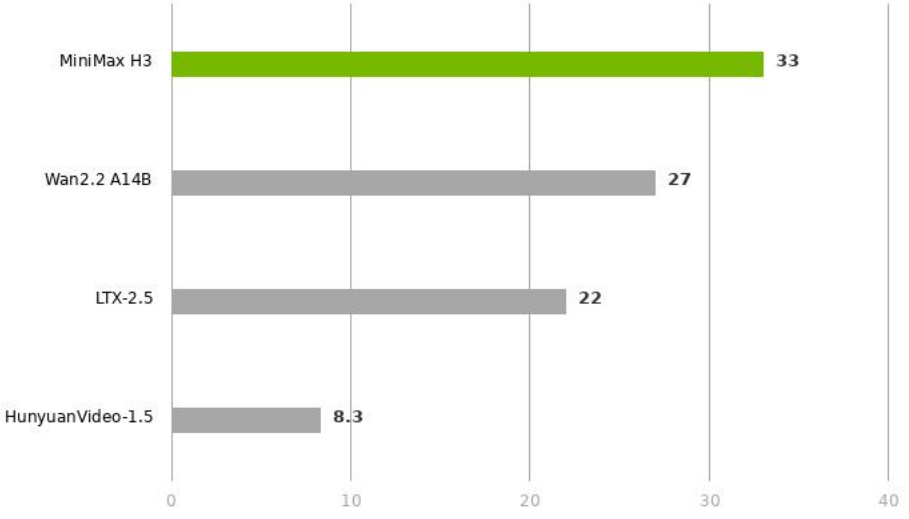


01 · Real-Time H3 Generation



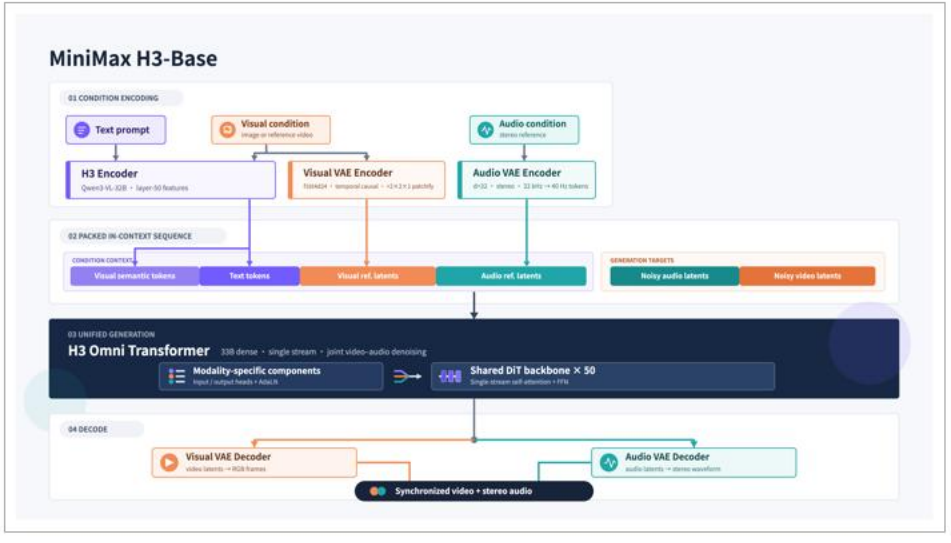
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

32B encoder → 33B dense Omni Transformer → visual/audio VAE. Fifty blocks × 50 steps keep compute and memory pressure high; about 13B AdaLN parameters can be cached

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: NIC → RDMA*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

Latency Breakdown

NCCL occupies 49.2% of the average GPU timeline

Stage	Average GPU Time*	Share
Mean GPU timeline*	250.0 s	100.0%
NCCL communication	123.0 s	49.2%
↳ DiT collectives	116.1 s	46.4%
↳ Encoder, VAE, and other collectives	7.0 s	2.8%
Tensor compute	84.5 s	33.8%
↳ Attention	45.3 s	18.1%
↳ GEMM / MLP	39.2 s	15.7%
AdaLN weight staging + device layout	38.5 s	15.4%
Encoder, VAE, and other GPU work	4.0 s	1.6%

- Of the 38.5 s average, 36.9 s is AdaLN projection weight staging from CPU; layout is 1.5 s
- This run uses the default runtime cache, not a complete precomputed schedule sidecar

Base Ref2VA · 960×544 · 124f @ 24 FPS · 50 steps · BF16 · 8× RTX PRO 5000 · default AdaLN offload · local NCCL / RDMA off · *sum across 8 GPU timelines ÷ 8

How Denoising Works

H3 follows a learned velocity field from noise toward clean video and audio latents

State and direction

- x(t) is the current latent; the DiT predicts a data-ward velocity

Solver update

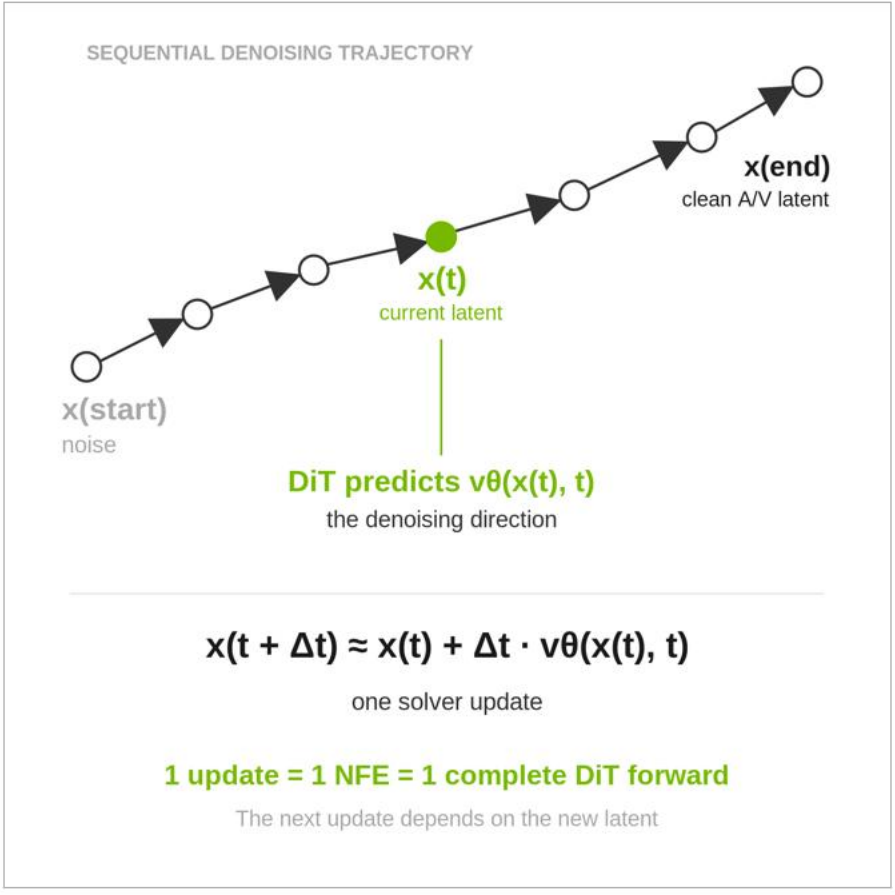
- Euler advances x(t) by a small interval along that velocity
- The next DiT call needs the new latent, so conventional steps run in sequence

Network function evaluation (NFE)

- One NFE is one complete DiT forward

Base H3

- 50 NFE × 50 main blocks = 2,500 main-block passes



Classical Flow-Matching Sampling

A chain of dependent solver updates turns noise into clean latents

Sources: MiniMax H3 Base configuration · NVIDIA FastGen-PDD sampling background (arXiv:2607.26004) · author abstraction

PDD Turns 50 DiT Passes Into 8

Predict four denoising directions in parallel; advance one interval block per pass

Teacher targets

- Base H3 supplies the mean denoising direction for each fine interval

One student pass

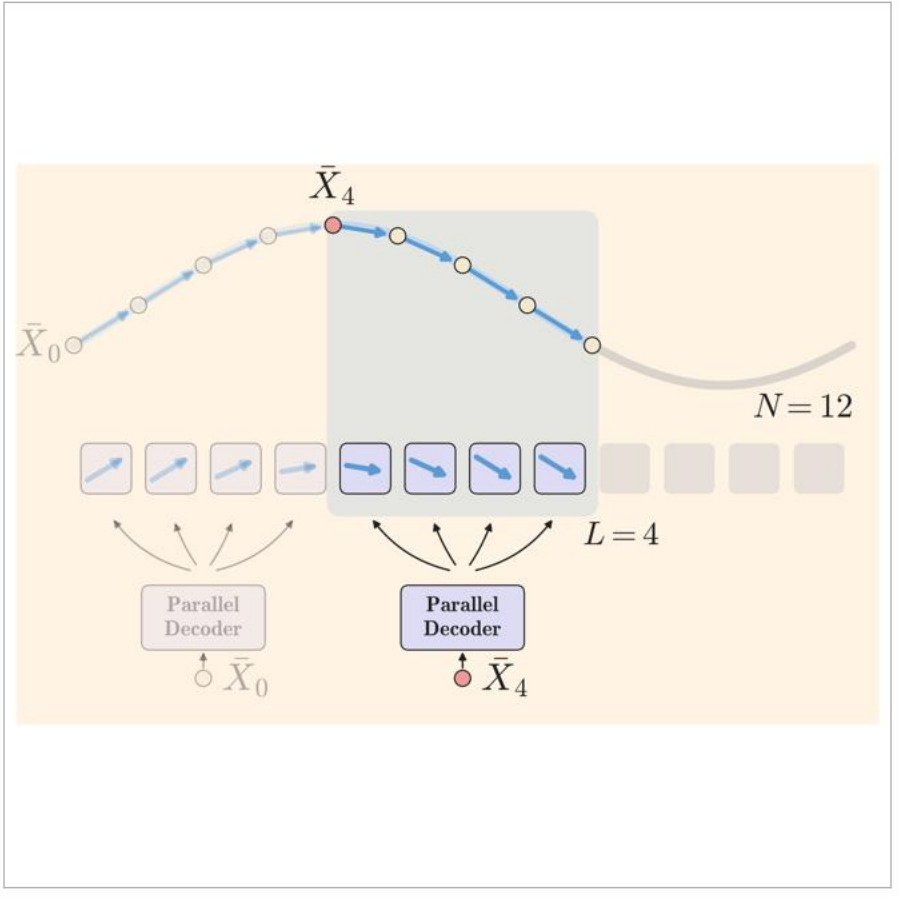
- From the same noisy latent, predict four consecutive directions in parallel

Advance and repeat

- Combine the four predictions to cross one block; repeat from the new latent

H3 PDD8

- Train a new 32-interval grid: 32 ÷ 4 = 8 passes
- Repeated block calls: 2,500 → 400 (−84%)



N = fine intervals · L = intervals predicted in parallel

Paper example: N = 12, L = 4 · H3 PDD8: N = 32, L = 4

Sources: NVIDIA FastGen-PDD, Fig. 2 (arXiv:2607.26004) · Alibaba-PAI MiniMax-H3 Ref2VA Acc-8Step @ 335001f

VSA Makes Attention Sparse

Video Sparse Attention learns which 3-D video cubes matter

Dense workload

- 34,373 tokens → 1.181B pairs/head/block at 544P

Learned VSA route

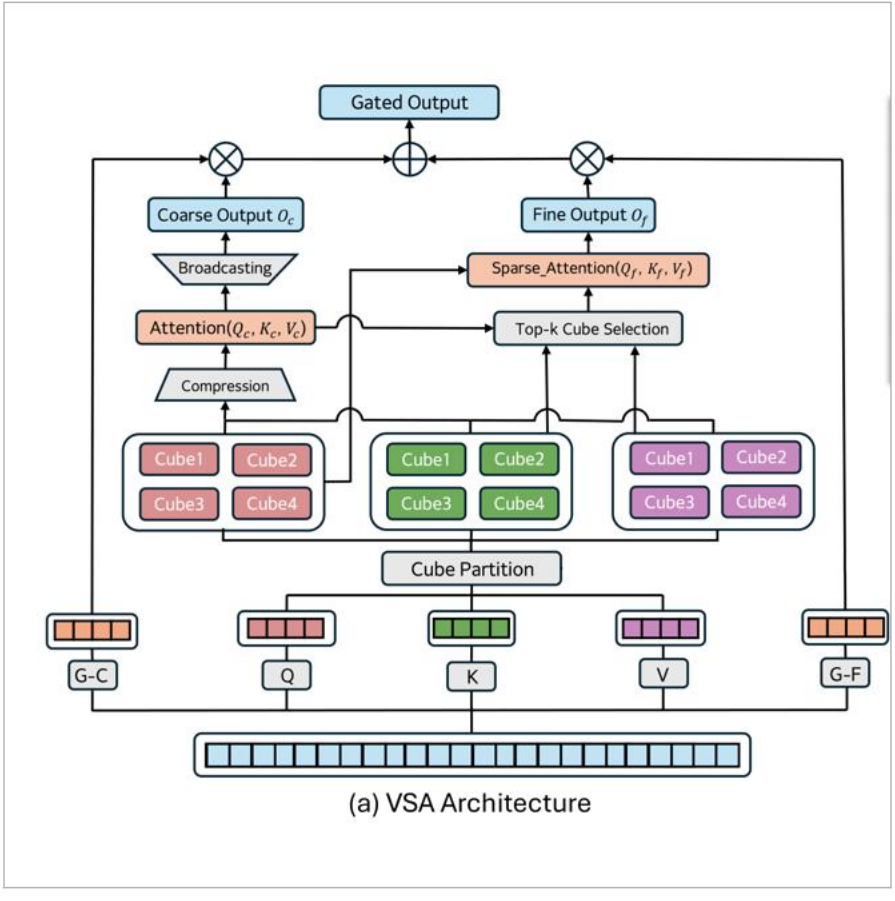
- Partition → coarse score → Top-K → fine attention → gate

Paper defaults

- 4 × 4 × 4 = 64 tokens per cube · Top-K = 32 KV cubes

System boundary

- Attention compute becomes sparse; USP8 still exchanges full-length Q/K/V/O
- Needs sparse-trained routing and gate weights



(a) VSA Architecture

Partition → Coarse → Top-K → Fine → Gate

Paper: 64-token cubes · Top-K = 32 KV cubes/query · H3: route 128 rows, kernel 64 rows

Source: Video Sparse Attention, arXiv:2505.13389, Fig. 1 · current H3 Ulysses communication path audited from source

QAD Quantizes Linears; Sage Quantizes VSA

Each path targets different SM120 kernels, and current evidence validates them separately

Path	Numerical Route	Effect	Validated Scope
QAD checkpoint	NVFP4 linears; BF16 exceptions	Recovers quantization error during distillation	8 forwards; PDD32; VSA 90%; tile 128
Sage VSA	Q/K INT8; V FP8; BF16 output	Quantizes tile64 fine attention after routing	SM120; d=128; opt-in; Ulysses
Precision island	Token refiner: BF16; VSA default: BF16	Keeps the sensitive short path in BF16	Same-seed quality check pending
Inter-GPU transport	BF16 Q/K/V before Sage; BF16 O after Sage	Keeps collective payloads in BF16	QAD and Sage do not reduce bytes

Sources: FastH3 Omni NVFP4-QAD rev 0087718 · NVIDIA ModelOpt QAD · vLLM-Omni SM120 Sage-VSA @ 63487a7

USP8 Issues Four All-to-Alls per DiT Block

Base50 runs 2,500 blocks at S = 34,432 on eight RTX PRO 5000 GPUs with local NCCL and no RDMA against a 5.167 s deadline

Scope	Tensor / Execution	A2A Rounds	Minimum Off-Rank Payload*	Rate at 5.167 s*
One DiT Block	Q / K / V forward + O reverse	4	1.728 GB	61.7 MB/rank/A2A
Base50	50 steps × 50 blocks = 2,500	10,000	4.319 TB	836.0 GB/s
PDD8	8 steps × 50 blocks = 400	1,600	691.1 GB	133.8 GB/s
Boundary	Current VSA / Sage keeps dense BF16 exchange	—	Logical bytes	Not achieved bandwidth

Source-derived @ vLLM-Omni H3 rev 63487a7f · logical one-way off-rank payload lower bound; not measured PCIe/NIC traffic or achieved bandwidth

CUDA Graph for Head Pipelining

All-head replay cuts launch gaps; head bands expose overlap

All heads + Graph

- Cuts launch overhead, but attention still waits for the full exchange

Head bands + Graph

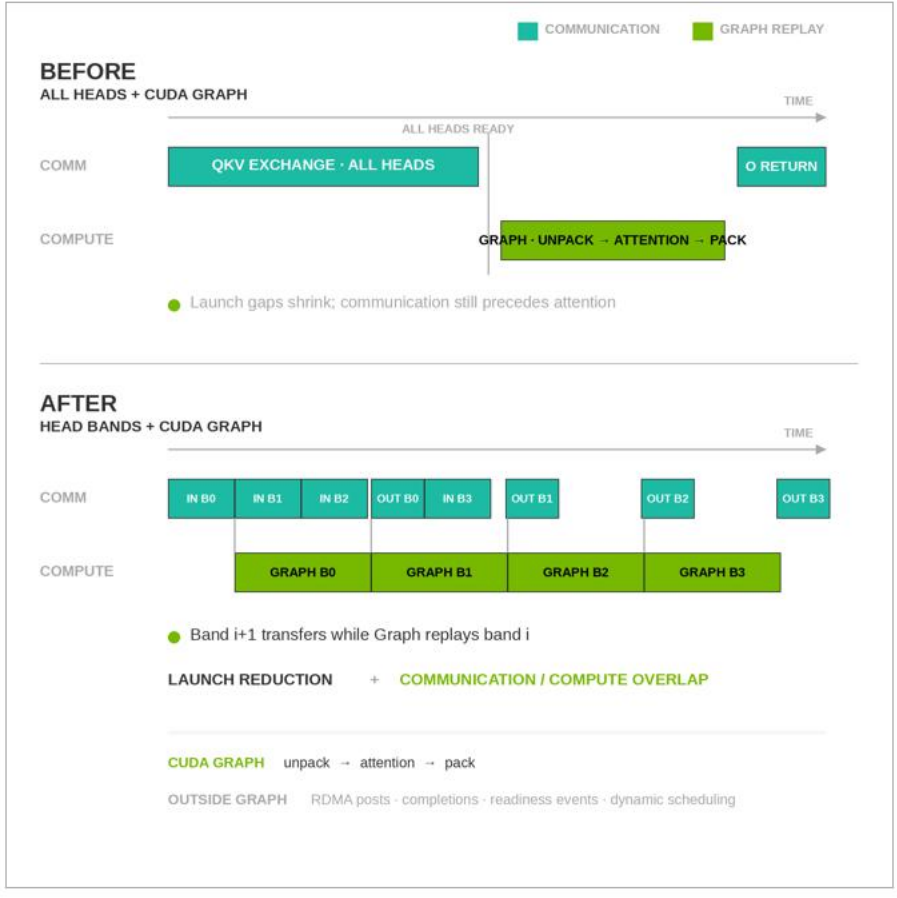
- Starts attention as each fixed band arrives and pipelines later bands

Measured compute-only

- All-head −0.39% · four bands −2.30%; banding pays off through overlap

Capture boundary

- Graph: unpack → attention → pack; RDMA and readiness events stay outside



Same Graph Benefit · Different Overlap

Measured compute-only: all-head −0.39% · four bands −2.30% · banding pays off through overlap

CUDA Graph Reuse Depends on the Video Profile

Arbitrary video requests have lower Graph reuse than LLM decode; fixed live streams reuse across continuation windows

Question	LLM Decode	H3 Video	Serving Policy
Shape key	Batch bucket q_len = 1	H/W/T · reference slots head band	Cache a small set of hot profiles
Replay window	Many token steps and requests	Across blocks + steps PDD8: 400 · Base50: 2,500*	Reuse persistent slots across continuation windows
VSA Top-K	—	Count fixed per profile indices and values change	Content changes replay the same Graph
New geometry	Select another batch bucket	Bucket hit or eager path	Capture only after the profile becomes hot
Buffer addresses	Static or pooled decode buffers	Registered QKV/O staging buffers	Use a persistent buffer pool

*Per band/profile GraphExec when configuration matches · (1,2,3,1) uses four GraphExecs · Sources: CUDA Programming Guide; current H3 prototype audit

Why This Request Uses USP8

DP for throughput · no CFG branch · USP8 for single-request latency

Single-request latency

- DP adds replicas and throughput; it does not split one request

No CFG branch

- H3 is CFG-distilled: one positive forward per step, so CFGP has nothing to parallelize

USP8 fits the work

- Eight ranks shard the 56,960-token sequence; Q/K/V/G/O cross the topology

Topology per rank

- 1 PXB · 2 NODE · 4 SYS peers per rank



DP1 / TP1 / USP8 / Ring1 · One Request, One Replica

Per rank: 1 PXB + 2 NODE + 4 SYS peers · every GPU has a nearest CX-8 exit

Sources: MiniMaxH3Pipeline and denoise-loop code audit · retained nvidia-smi topo -m · 8 nearest 400-Gb/s CX-8 ports

GPUDirect RDMA Shortens the Route

960×544 · 362f · BF16 · DP1 / TP1 / USP8 / Ring1 · S = 56,960 · Q/K/V/G/O

Route	Logical Off-Rank Volume	Rank-Max / Block	Effective BW / Rank	Serialized PDD8*
NCCL local	446.6 MB / rank / block 1.429 TB / all ranks / PDD8 request	24.87 ms	17.95 GB/s	9.95 s
Nearest-NIC GDR	446.6 MB / rank / block 1.429 TB / all ranks / PDD8 request	10.86 ms	41.14 GB/s	4.34 s
RDMA delta	Same five-slot bytes	−56.4%	2.29×	−5.61 s
Measured scope	Communication + path-required layout	Exact vs independent all_gather	No attention or GEMM	No overlap *serial × 400; not E2E

Measured 2026-10-25 · 5 warmups + 8 trials × 4 calls · rank-max then mean · *serialized block mean × 400; not end-to-end

RDMA Precheck Passes 96/96

Four GPUs are byte-exact; eight-GPU H3 remains pending

Q/K first

- V waits for an explicit CUDA event

Graph replay

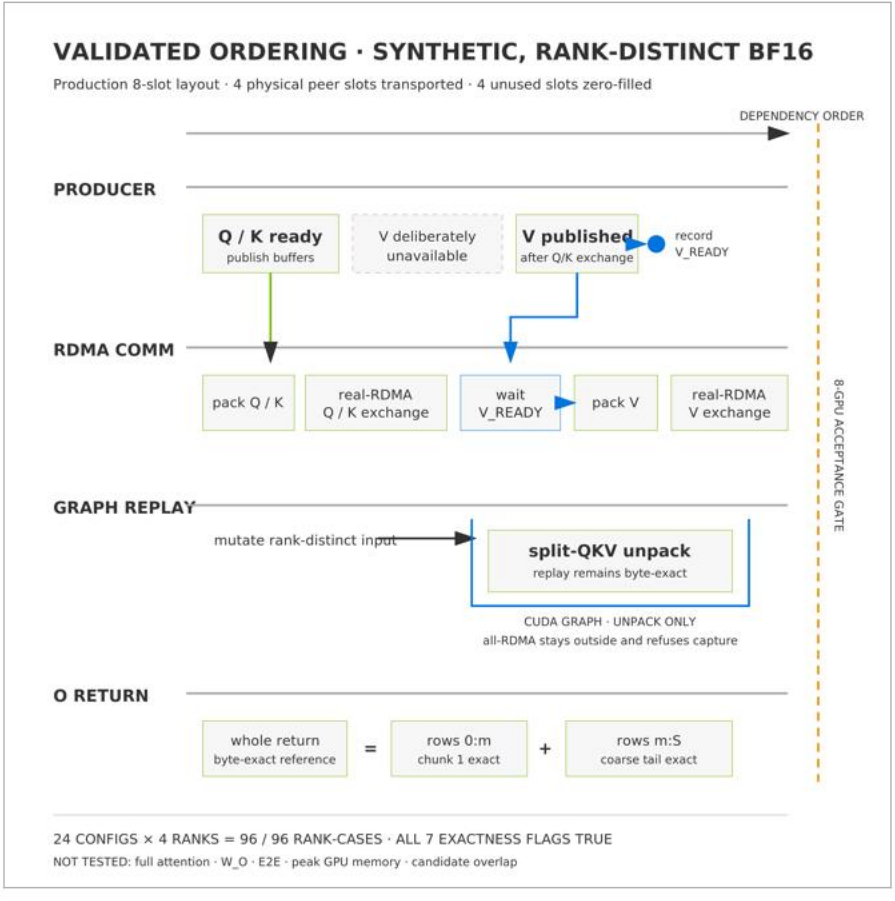
- Fixed-address unpack remains exact after the input changes

Tiled O return

- Whole and row-chunk returns match byte for byte

Eight-GPU acceptance gate

- Full attention + W_O, end-to-end latency, memory, and Nsight overlap



Four-GPU Real RDMA · 96 of 96 Rank Cases Exact

24 configs × 4 ranks · 8-GPU H3 candidate pending

Measured evidence: real-rdma-four-gpu-20261024-v2 · 2026-10-24 · candidate correctness precheck; latency and overlap not measured

Real-Time Demonstrated; Attribution Pending

Only Base50 uses the locked 960×544, 124-frame request; each optimized result uses a different workload

Status	Workload	System	Result / Limit
Measured baseline	960×544 · 124f · 5.167 s	Base50 · BF16 · USP8 · NCCL	247.956 s · RTF 47.992
15 s study	960×544 · 362f · 15.083 s	PDD8 · QAD · VSA · RDMA	14.974 s mean · RTF 0.993
10 s study	960×544 · 243f · 10.125 s	4-step Turbo · MXFP8 · Sage · RDMA	6.958 s · RTF 0.687
Portrait study	576×1024 · 119 new frames · 4.958 s	4-step MXFP8 · Sage · FP4 VAE · RDMA	4.459 s median · RTF 0.899
Required matched test	Lock image, audio, prompt, seed, shape, and request mode	PDD8 → VSA → QAD/Sage → RDMA	At least 3 runs per stage

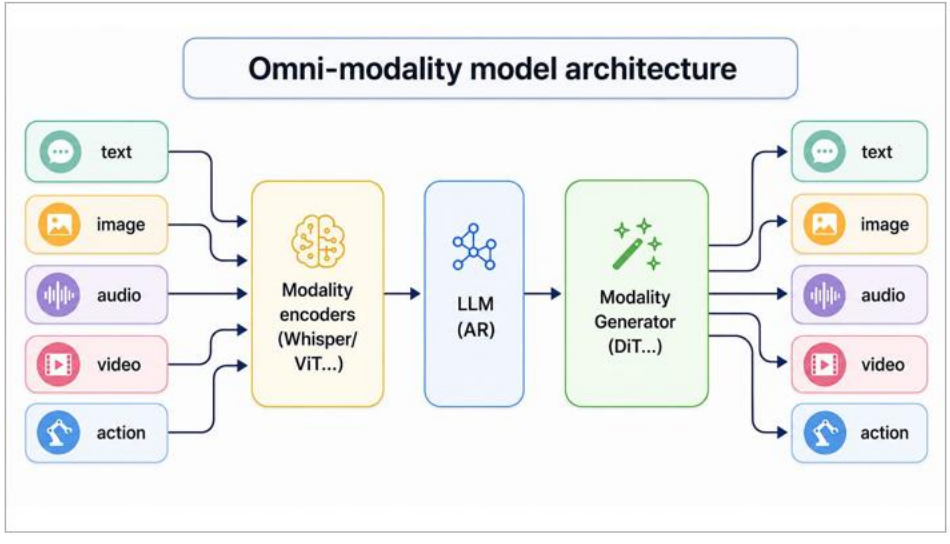
Sources: [A] Base50 r4 · [B] QAD 15 s cohort · [C] paired 10 s study · [D] portrait production · compare only within a row; timing and memory methods differ

02 · ComfyUI Apps With vLLM-Omni



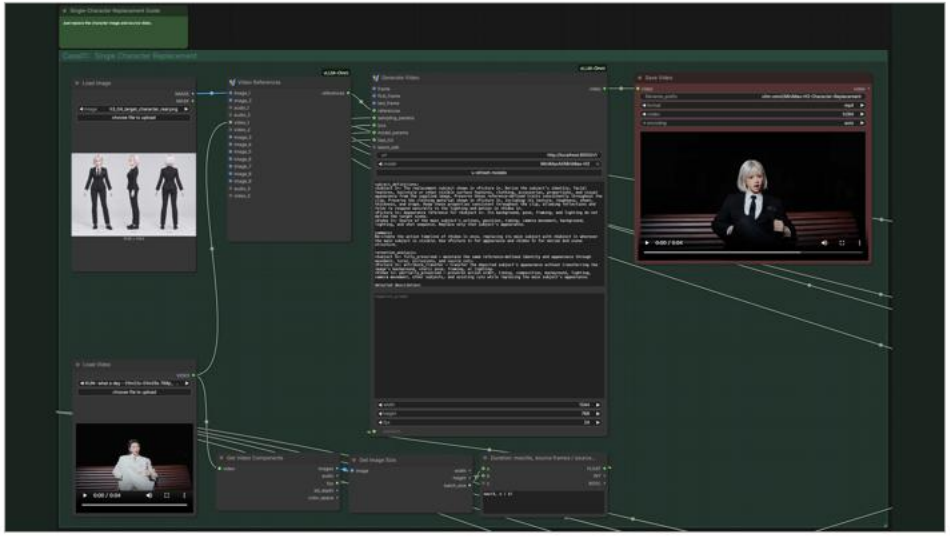
vLLM-Omni Brings Multi-GPU H3 to ComfyUI

Open PR #9194 adds character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI authors the workflow; vLLM-Omni runs scheduling, model execution, and multi-GPU deployment



ComfyUI Workflow · PR #9194

The workflow supports single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

